PROJECT REPORT

CREDIT CARD FINANCIAL BEHAVIOR & RISK ANALYSIS

Prepared By:

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Tools Used:

SQL | Excel | Power BI

Year:

2026

EXECUTIVE SUMMARY

This project analyzes credit card customer data to identify financial behavior patterns and assess default risk. Using a dataset of 30,000 customers, the analysis focuses on repayment history, credit utilization, and demographic factors.

The objective is to identify high-risk customers and generate insights that can support financial institutions in reducing defaults and improving credit decision-making.

The analysis reveals that customers with high credit utilization and delayed payments are significantly more likely to default. Approximately 22% of customers fall into the high-risk category, indicating a critical need for monitoring and intervention strategies.

PROBLEM STATEMENT & OBJECTIVES

Problem Statement

Credit card defaults lead to significant financial losses. Identifying risky customers early can help institutions reduce bad debt and improve profitability.

Objectives

- Identify customers with high default risk
- Analyze repayment behavior
- Study credit utilization patterns
- Segment customers based on risk
- Provide business recommendations

DATASET DESCRIPTION

The dataset contains detailed information about credit card customers, including financial and demographic data.

Key Features

- Credit Limit (LIMIT_BAL)
- Gender (SEX)
- Education Level
- Marriage Status
- Bill Amount (Last 6 Months)
- Payment Amount (Last 6 Months)
- Repayment Status
- Default Indicator

Key Columns

- LIMIT_BAL (Credit Limit)
- SEX (Gender)
- EDUCATION
- MARRIAGE
- BILL_AMT1–6
- PAY_AMT1–6
- PAY_0–PAY_6
- default.payment.next.month

Dataset Size

- Total Records: 30,000
- Total Columns: 23

DATA CLEANING & PREPARATION

Data preprocessing was performed using Excel and SQL.

Steps Performed

- Removed inconsistencies and null values
- Converted categorical codes into readable values
- Standardized column names
- Created derived features:
    - Total Bill Amount
    - Total Payment Amount
    - Credit Utilization Ratio
- Handled outliers in credit limits

This ensured high-quality and reliable data for analysis.

SQL ANALYSIS

SQL was used to extract key insights from the dataset.

Analysis Performed

Query 1: Total Customers

```
SELECT COUNT(*) AS total_customers

FROM credit_card;
```

Query 2: Default Rate

```
SELECT

  COUNT(CASE WHEN default_payment_next_month = 1 THEN 1 END) * 100.0 / COUNT(*) AS default_rate

FROM credit_card;
```

Insight

Out of 30,000 customers, approximately **22% are likely to default**, indicating a significant risk segment.

Advanced Analysis

Query 3: High-Risk Customers (Payment Delay)

```
SELECT COUNT(*) AS high_risk_customers

FROM credit_card

WHERE PAY_0 > 2;
```

Query 4: Credit Utilization

```
SELECT

  AVG(BILL_AMT1 / LIMIT_BAL) AS avg_utilization

FROM credit_card;
```

Insight

Customers with **high utilization ratios and repeated delays** show a strong likelihood of default, making them critical for monitoring.

POWER BI DASHBOARD OVERVIEW

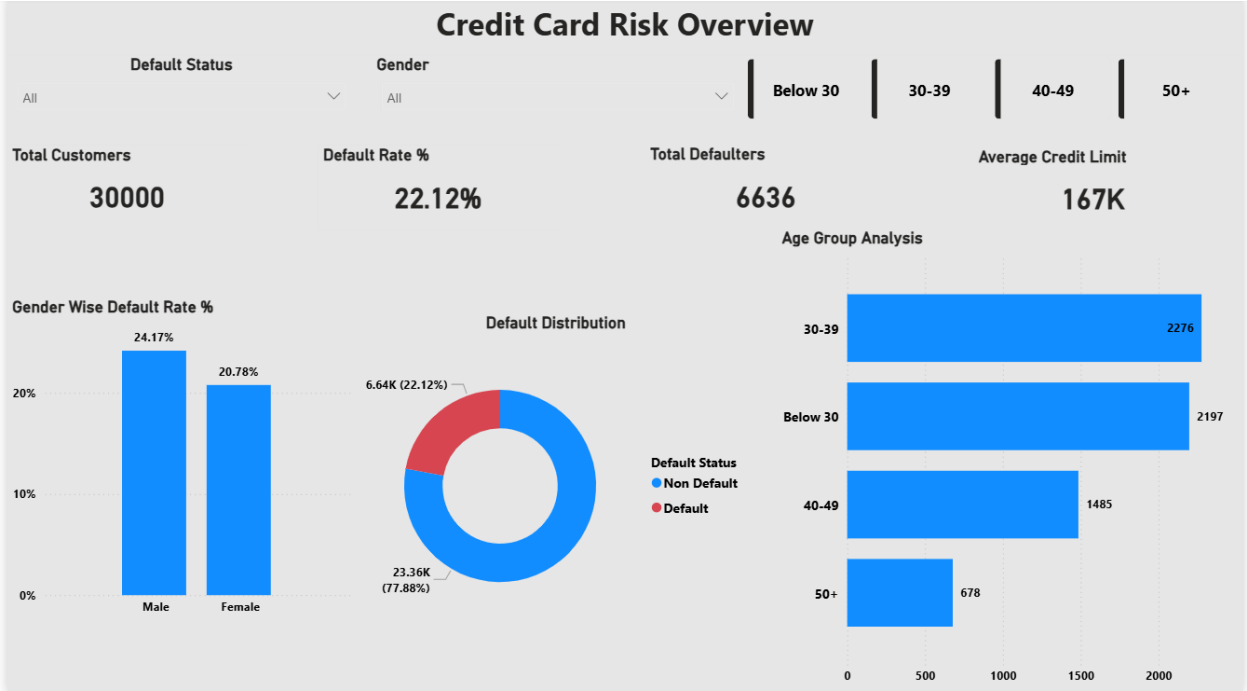
The Overview dashboard provides a high-level summary of customer data.

Key Visuals

- Total Customers: 30,000
- Default Rate: 22.12%
- Average Credit Limit: 167K

Insight

Approximately 22.12% of customers are at risk of default, meaning nearly 1 out of every 5 customers may fail to repay. The default distribution chart shows a significant proportion of risky customers, while age group analysis indicates that customers aged 30–40 contribute the highest number of defaulters. Additionally, gender-based analysis shows slightly higher default rates among males compared to females.



POWER BI DASHBOARD – RISK ANALYSIS PAGE

This page focuses on identifying high-risk customers.

Key Visuals

- Default Rate by Age Group
- Default Rate by Education
- Payment Delay Analysis

Insight

• Customers in the 30–40 and 50+ age groups show relatively higher default rates compared to younger segments, indicating increased financial risk in these categories.

• Education-based analysis shows that customers with lower or unspecified education levels tend to have higher default rates, suggesting a possible link between financial awareness and repayment behavior.

• Marital status shows moderate variation, with similar risk levels across categories, indicating it is a less significant factor in predicting default.

• Payment delay patterns clearly indicate that customers with repeated delays are significantly more likely to default, making payment behavior the strongest risk indicator.

POWER BI DASHBOARD – BEHAVIOR ANALYSIS PAGE

This page analyzes customer financial behavior.

Key Visuals

- Credit Utilization by Risk Segment
- Bill vs Payment Comparison
- Default Rate by Delayed Months

Insight

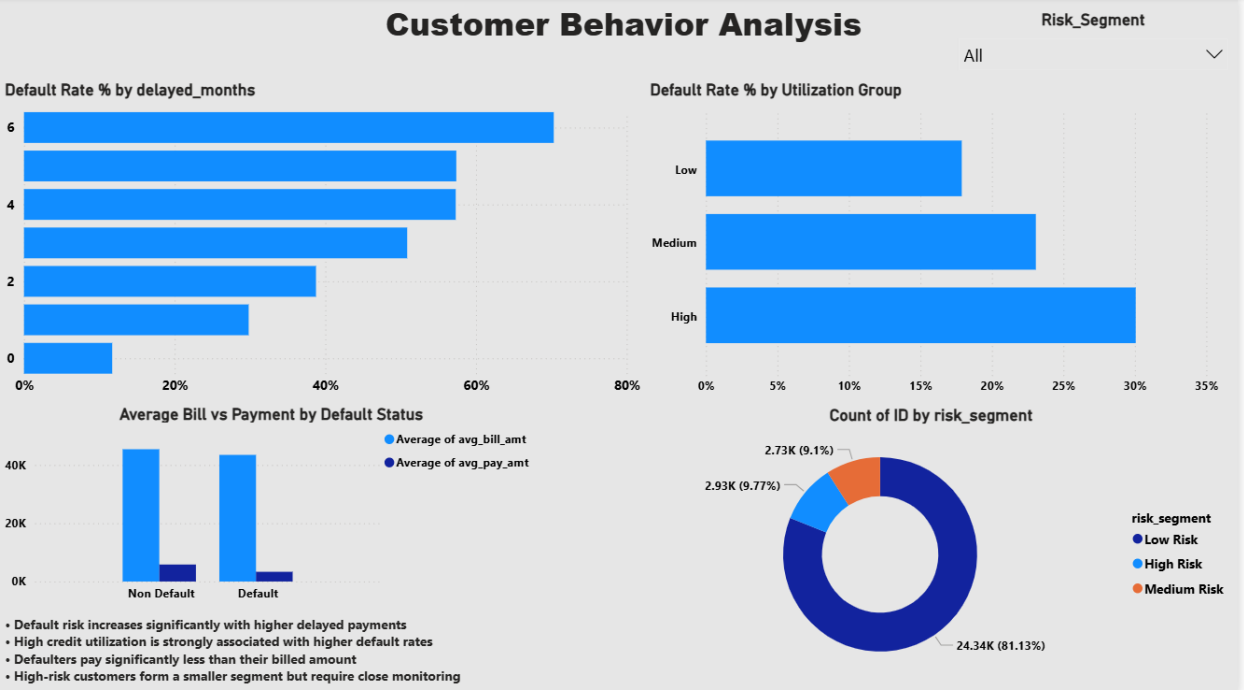
• Customers with high credit utilization exhibit significantly higher default rates, with risk levels exceeding 30%, indicating that over-dependence on credit is a major risk factor.

• Analysis of delayed payments shows a clear upward trend in default probability, where customers with multiple months of delay are substantially more likely to default.

• The bill vs payment comparison highlights that defaulters consistently pay significantly less than their billed amount, reflecting poor repayment discipline.

• Risk segmentation reveals that a majority of customers fall under the low-risk category, but a smaller segment of high-risk customers contributes disproportionately to default cases.

• Overall, financial behavior indicators such as credit utilization and payment consistency are stronger predictors of default risk than demographic factors.



KEY INSIGHTS & BUSINESS RECOMMENDATIONS

Key Insights

- High credit utilization strongly increases default risk
- Payment delays are the strongest indicator of financial stress
- Consistent repayment behavior reduces default probability

Business Recommendations

- Monitor customers with high utilization ratios
- Implement stricter policies for delayed payments
- Offer incentives for timely repayment
- Segment customers for targeted credit strategies

Business Impact

These insights can help financial institutions reduce default risk, improve credit scoring, and optimize customer targeting strategies.

CONCLUSION & SKILLS USED

Conclusion

The project successfully identifies high-risk customers and highlights important behavioral patterns affecting credit default.

Skills Used

- SQL for data analysis
- Excel for data preprocessing
- Power BI for visualization

Project Outcome

- Built a 3-page interactive dashboard
- Identified key risk indicators
- Delivered actionable insights for financial decision-making

Final Statement

This project demonstrates my ability to analyze real-world financial data, perform SQL-based analysis, and build interactive dashboards to support data-driven business decisions.